

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 2025
Project Name	ServiceNow-Based Employee Laptop Request Application

Technical Architecture:

The proposed solution is implemented on the ServiceNow cloud platform. Employees submit laptop requests through the ServiceNow interface. The request is stored in the ServiceNow database, duplicate validation is handled by a Business Rule, and approval is automated through Flow Designer. Notifications are sent to the appropriate users. Access is controlled using roles and ACLs.

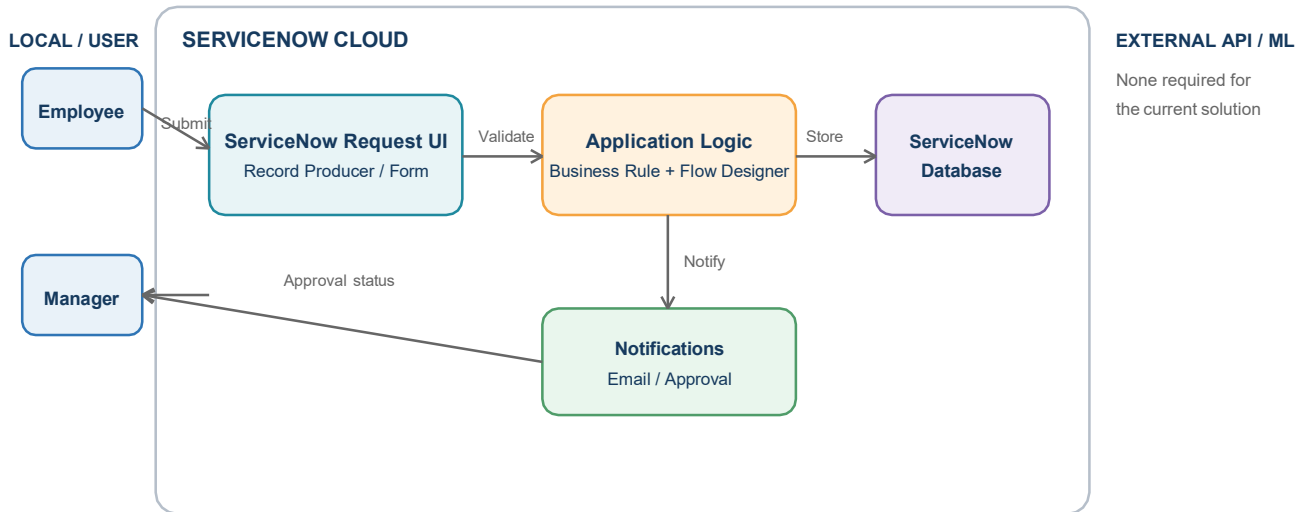


Table-1: Components & Technologies

1	User Interface	Employee laptop request and approval interaction.	ServiceNow UI / Record Producer
2	Application Logic-1	Checks for duplicate employee laptop requests.	ServiceNow Business Rule
3	Application Logic-2	Routes request for manager approval and updates status.	ServiceNow Flow Designer
4	Application Logic-3	Controls user access by role and permissions.	ServiceNow Roles + ACLs
5	Database	Stores laptop request records and related fields.	ServiceNow platform database
6	Cloud Database	Cloud-hosted platform data storage.	ServiceNow Cloud
7	File Storage	No separate file storage is required in the current scope.	Not applicable
8	External API-1	No third-party API is required for the current workflow.	Not applicable
9	External API-2	No second external API is required.	Not applicable
10	Machine Learning Model	No ML model is required for request, approval or duplicate validation.	Not applicable
11	Infrastructure	Application deployment and execution environment.	ServiceNow Cloud Platform

Table-2: Application Characteristics

1	Open-Source Frameworks	The solution is primarily configured using ServiceNow platform capabilities rather than a separate open-source application framework.	ServiceNow platform configuration
2	Security	Role-based access and ACLs restrict who can create, read, write and manage request records.	ServiceNow Roles + ACLs
3	Scalable Architecture	Cloud platform architecture supports centralized application services and data management as request volume grows.	ServiceNow Cloud
4	Availability	The application is hosted on the ServiceNow cloud platform rather than a local server.	ServiceNow Cloud
5	Performance	The workflow uses platform-native forms, Business Rules and Flow Designer to reduce manual processing and provide timely status updates.	ServiceNow platform services